



AN INTELLIGENT URBAN PARKING MANAGEMENT FRAMEWORK INTEGRATING DYNAMIC PRICING AND GEOSPATIAL RESOURCE OPTIMIZATION

Presented by:

Joshua Arnold Leo - 727823TUCS122

Jeswanth A - 727823TUCS121

Harish C S - 727824TUCS701

Guided by:

Dr. K A Jayabalaji
Assistant Professor



ABSTRACT

Drivers spend significant time searching for available parking spaces, contributing to traffic, fuel waste, and emissions.

This project proposes an Urban Smart Parking System that integrates:

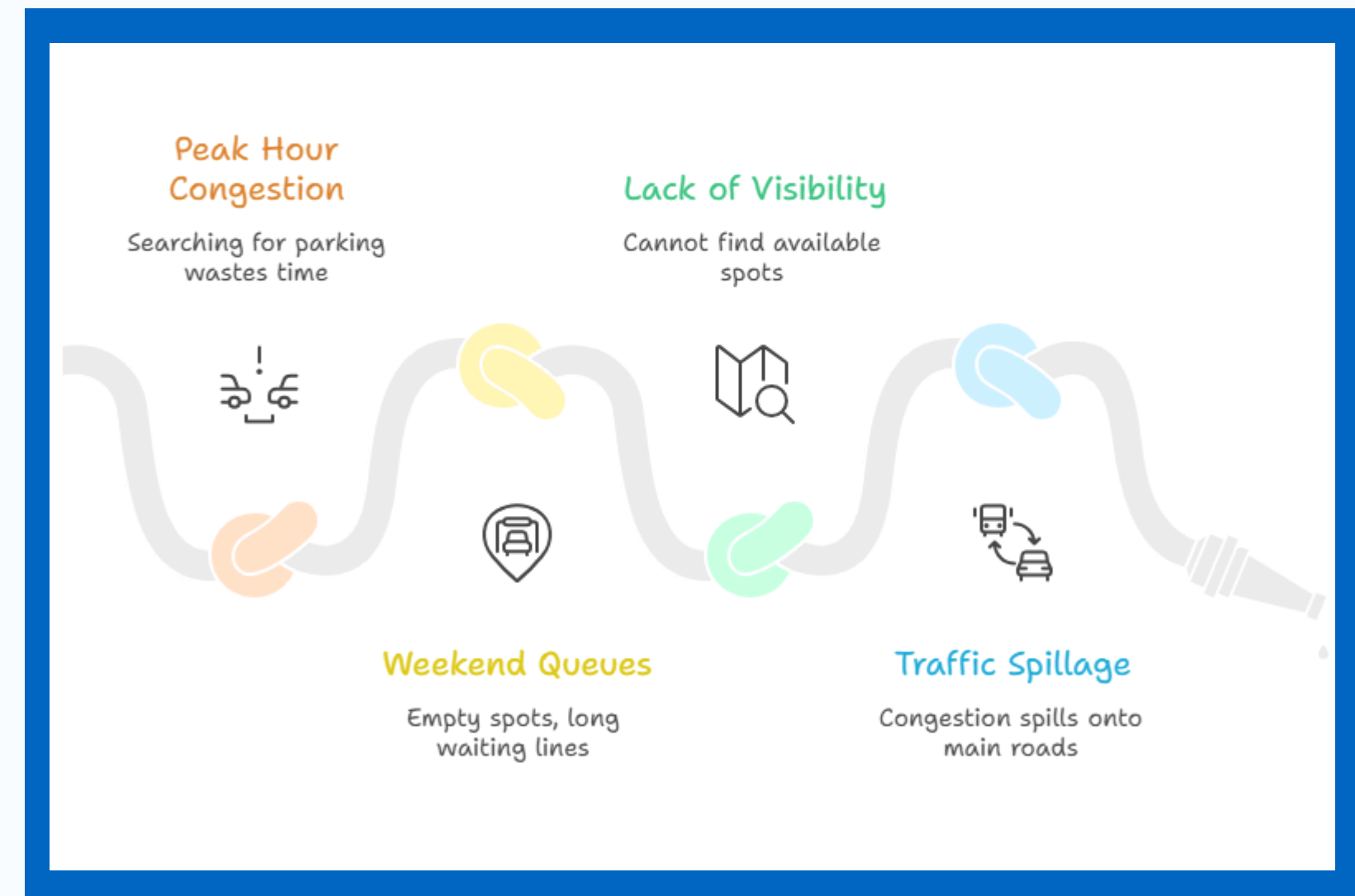
- Real-time parking availability
- Dynamic pricing based on demand
- Intelligent parking recommendation
- Role-based dashboards for drivers, managers, and authorities



PROBLEM STATEMENT

Urban cities suffer from:

- 🚗 25–30 minutes average parking search time.
- 📍 Lack of real-time space-level availability.
- 🚦 Increased congestion due to parking circulation.
- 💰 Static pricing leading to uneven space utilization.
- 📊 No centralized analytics for city planning



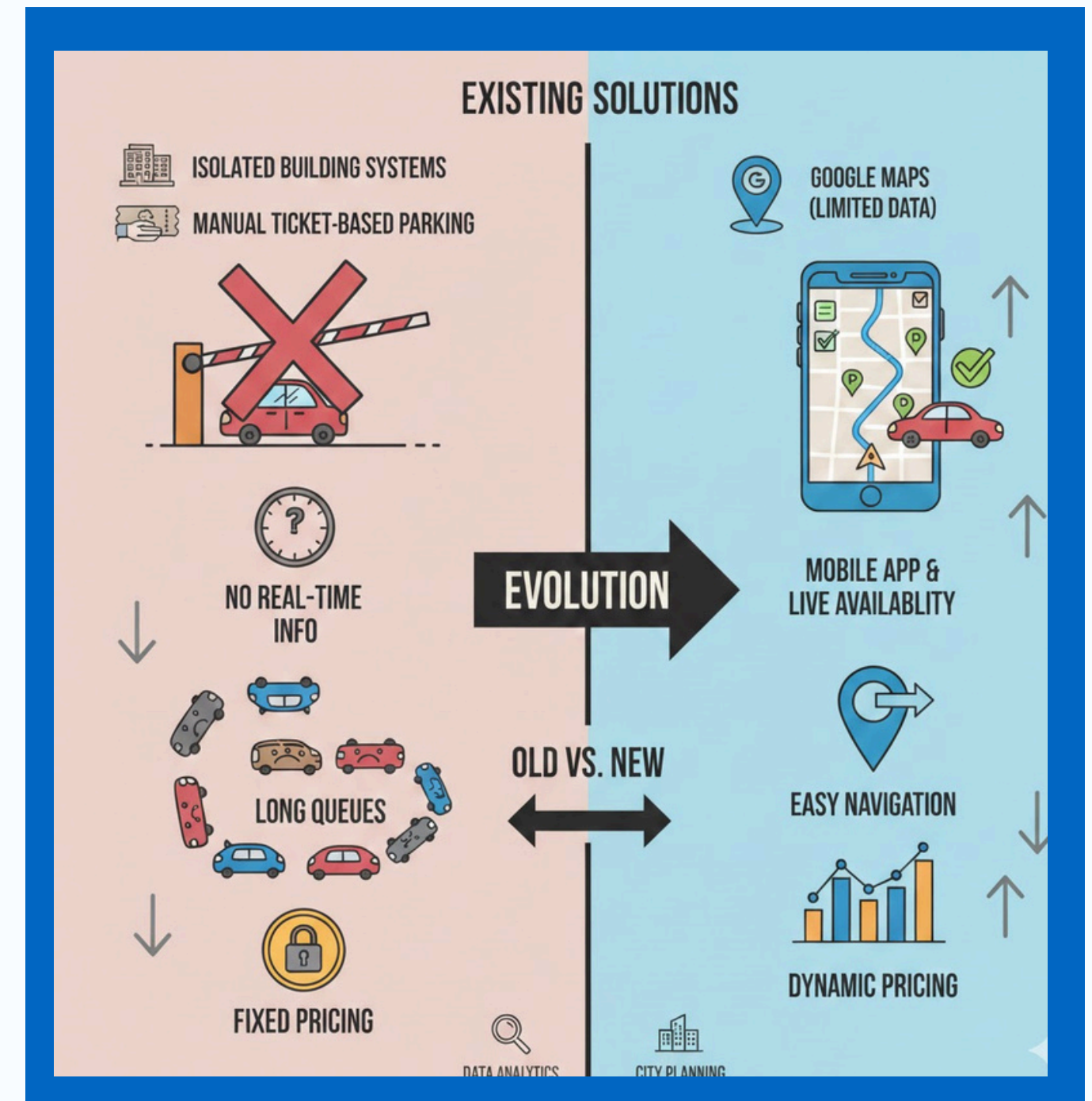
EXISTING SOLUTION

Current Solutions:

- Google Maps (limited parking metadata)
- Individual mall/building parking systems
- Manual ticket-based parking
- Sensor-based isolated systems.

Gaps Identified:

- No unified city-wide platform
- No intelligent recommendation engine
- No demand-based pricing
- No multi-role analytics dashboards
- Limited integration with traffic systems.



PROPOSED SOLUTION

A centralized smart parking platform that:

- Displays real-time parking availability on a city map.
- Guides drivers to nearest and best parking option.
- Applies dynamic pricing based on demand.
- Provides analytics dashboards for managers and authorities.

Enhancement Over Base Papers:

- Goes beyond space detection
- Adds pricing intelligence
- Integrates route optimization
- Supports city-wide analytics



TECH STACK

Frontend:

- React.js
- Leaflet.js (Map visualization)
- Chart.js (Analytics dashboard)

AI/Logic:

- Rule-based dynamic pricing algorithm
- Nearest-neighbor parking recommendation
- Time-series occupancy trend analysis.

Backend:

- Spring Boot (REST APIs)
- WebSocket (Real-time updates)
- Spring Security (Role-based access)

External Services:

- OpenStreetMap (Map tiles)
- OSRM (Routing engine)

Database: MySQL



ADVANTAGES

Advantages:

1. Decreases average search time by up to 70–80% through real-time availability.
2. Optimizes parking utilization and balances congestion across zones.
3. Suggests best parking based on distance, availability, cost, and user preferences.
4. Designed for multi-lot and multi-city deployment with geospatial support.
5. Dynamic pricing increases fairness and boosts municipal income.



FUTURE SCOPE

Future Scope:

1. Integration of CNN-based real-time vision models.
2. Machine learning models for accurate occupancy forecasting.
3. Real IoT-based slot detection for higher accuracy.
4. Automatic Number Plate Recognition (ANPR) for seamless permit.
5. Expansion into integrated smart mobility platforms.

REFERENCES

1. **iParker—A New Smart Car-Parking System Based on Dynamic Resource Allocation and Pricing(IEEE):**

- Amir O. Kotb, Yao-Chun Shen, Xu Zhu, Senior Member, IEEE, and Yi Huang, Senior Member, IEEE

2. **Smart Parking System with Computer Vision(IRJAEH):**

- Department of Artificial Intelligence and Data Science, Erode Sengunthar Engineering College, Erode, Tamil Nadu, India.



SYSTEM WORKFLOW

Parking App Workflow: From Search to Insights



User Opens App

User launches the parking application.

Searches for Parking

Enters location, date, and time preferences.

System Checks Availability

Database is queried for real-time parking space status.

Recommends Best Parking

System analyzes distance, availability, and pricing to suggest optimal options.

Navigation to Parking

User receives route guidance to selected parking lot.

Admin Dashboard Updates

System updates analytics dashboards with occupancy and usage insights.

CONCLUSION

The proposed system:

- Reduces parking search time by up to 80%
- Decreases urban congestion by optimizing distribution
- Improves revenue through dynamic pricing
- Enables data-driven city planning.

Academic Contribution:

- Integrates computer vision research concepts
- Extends them into a scalable smart city platform
- Combines AI, geospatial intelligence, and system design





THANK YOU